

**Mobile Application Development (MAD)**  
CCIT 4059, 2023-2024 Semester 2  
Final Project Submission Form

|               |              |                  |                    |             |                 |
|---------------|--------------|------------------|--------------------|-------------|-----------------|
| Class:<br>No. | <b>CL-02</b> | Student<br>Name: | <b>Ho Sze Tung</b> | Student ID. | <b>20201806</b> |
|---------------|--------------|------------------|--------------------|-------------|-----------------|

|                    |                          |              |                                                                                     |
|--------------------|--------------------------|--------------|-------------------------------------------------------------------------------------|
| Presentation Order | Lesson: 11 (A)Order: #16 | App.<br>Icon |  |
| Application Title: | Healthy Buddy            |              |                                                                                     |

- \* Highlight those different or extra parts from Progress Submission Form.**
- \* You may attach more details as an appendix if given space is not enough.**
- \* Ensure all the files can be open (no corruption) before submission**
- \* No Late Submission is accepted. No other means (such as email) of re-submission)**

**1) Items of submission (Checklist) and general evaluation:**

**Softcopies (Submit to SOUL)**

☐ Complete Program Source:

| Item | Softcopies of project files and File names for submission on SOUL (App Files)                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>HO_06_madA.apk</b>                                                                                                                           |
| 2    | <b>HO_06_madA.zip</b>                                                                                                                           |
| 3    | <b>HO_06_madA.mp4</b>                                                                                                                           |
| 4    | <b>readme_HO_06_madA.txt</b>                                                                                                                    |
| Item | Softcopies of project files and File names for submission on SOUL (Resources Files)                                                             |
| 1    | <b>app_icon.png</b>                                                                                                                             |
| 2    | <b>Background image: finishbg.jpg, homebg.bg, methodbg.webp, settingbg.jpg, sportbg.jpg, startbg.jpg, storybg.webp, trainingbg.jpg, usb.jpg</b> |
| 3    | <b>ImageView: buddyhi.png, lung.png, storyimg.png, using.png, welldone.png</b>                                                                  |
| 4    | <b>Sound: bgmusic.mp3</b>                                                                                                                       |

☐ Project Documentation (word doc and pdf files):

| Item | File name for submission on SOUL |
|------|----------------------------------|
| 1    | <b>MAD_presentation.pptx</b>     |
| 2    | <b>MAD_presentation.pdf</b>      |
| 3    | <b>HO_06_final.docx</b>          |
| 4    | <b>HO_06_final.pdf</b>           |

## HardCopies (Color Printing is NOT required)

☐ Project Submission Form (Hard Copy) (double sides)

☐ Presentation Slides, multiple slides printed in one page (4 in 1) and double sides

## Basic Evaluation

### Technical Requirement, Compilation & Operation

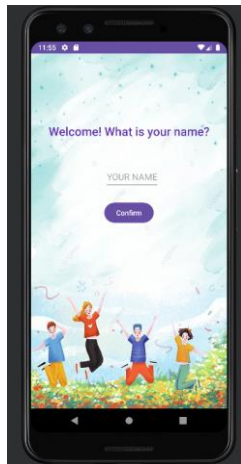
\* Description should be given (in remarks and/or later sections) in case of not 100% completed.

| Items of Requirements                                    | Self-Evaluation  |         |         |                       |                       |  | Remarks                                                      |
|----------------------------------------------------------|------------------|---------|---------|-----------------------|-----------------------|--|--------------------------------------------------------------|
|                                                          | 0%<br>(Not Done) | 25<br>% | 50<br>% | 75<br>%               | 100%<br>(Completed)   |  |                                                              |
| Multiple-Screens (Windows), with proper interconnections |                  |         |         |                       | <input type="radio"/> |  |                                                              |
| Story Behind & About Us sections                         |                  |         |         |                       | <input type="radio"/> |  |                                                              |
| (Basic Media) Image and Sound                            |                  |         |         |                       | <input type="radio"/> |  |                                                              |
| (Basic Input) e.g. One of Touch, Gesture, Accelerometer  |                  |         |         |                       | <input type="radio"/> |  |                                                              |
| [Self-learning techniques or methods]                    |                  |         |         |                       | <input type="radio"/> |  |                                                              |
| Compilation errors & warnings                            |                  |         |         | <input type="radio"/> |                       |  | No fatal Error, but MainActivity.java have 11 yellow warning |
| Major Run-time errors during execution                   |                  |         |         |                       | <input type="radio"/> |  | No                                                           |

## 2) Details of Design and Implementation

|                       |               |  |  |
|-----------------------|---------------|--|--|
| A) Application Title: | Healthy Buddy |  |  |
|-----------------------|---------------|--|--|

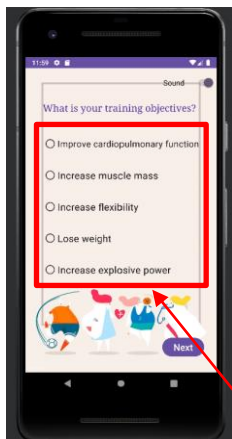
Home page (If first launch):



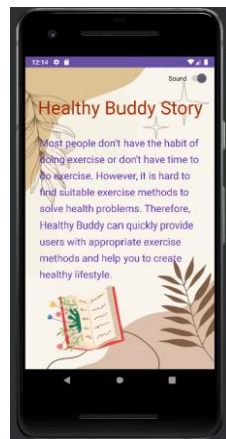
Main page:



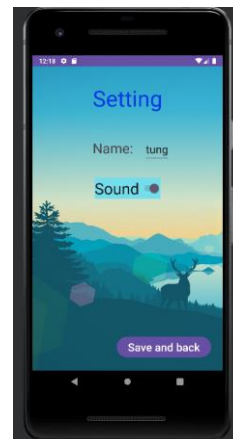
Start page:



Story page:



Setting page:

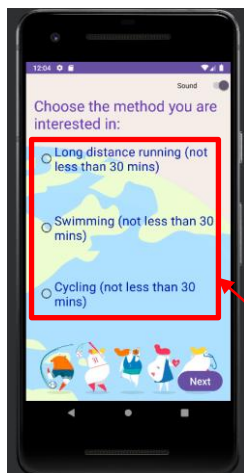


About Us page:

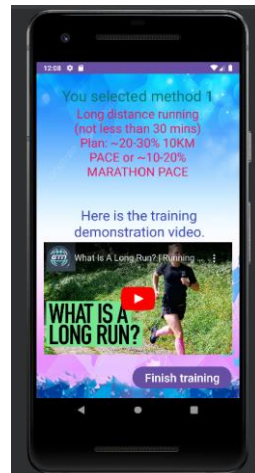


Select option

Method page (If choose option 1):



Training page (If choose method 1):



Watch demonstration video for training

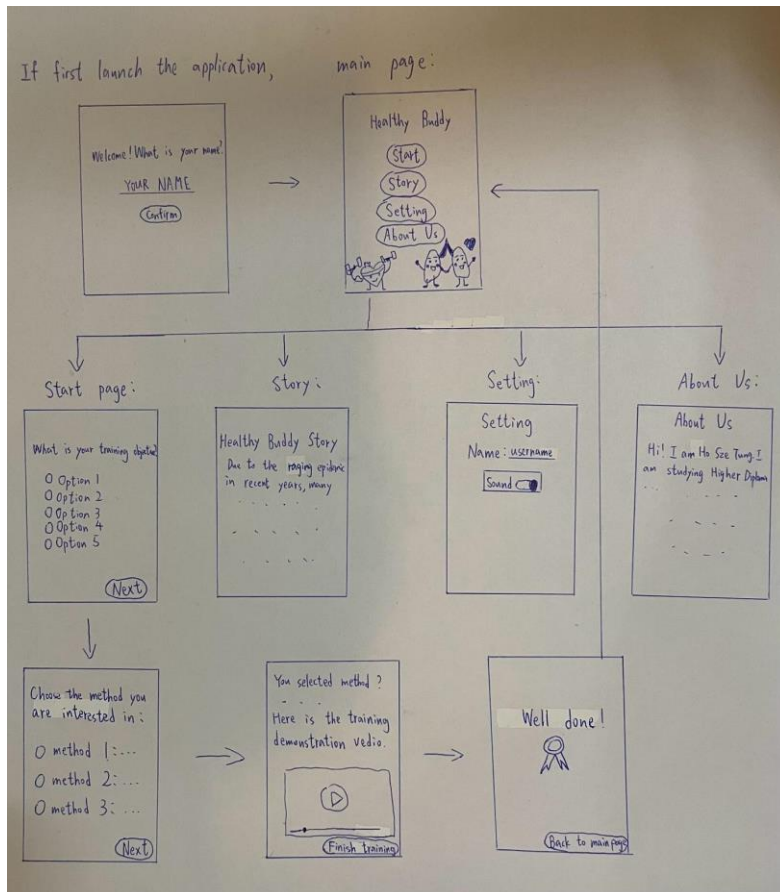
Finish page:



## B) Objectives, Target Users, and General Design Ideas

- The major purpose of this application is to quickly provide users with appropriate exercise methods to address health problems and create the habit of doing exercise.
- The objectives include:
  - a. To quickly provide users with appropriate exercise methods
  - b. Increase motivation to do exercise
  - c. Provide healthy lifestyle
- Target users are mainly:
  - a. People who like sports
  - b. People who are underweight, overweight or obesity
  - c. Busy people
- While some similar mobile apps are already in the market, our project focuses the convenience and efficiency to provide training method to user.
- General Design Ideas & Styles:
  - a. To quickly provide users with appropriate exercise methods by choosing training objectives
  - b. Provides training demonstration videos for user
  - c. General style: simple, comfortable, relaxed with two cartoon characters

**C) Design Details** (Include **Hand-Drawn storyboards , Prototypes** – scan them here)



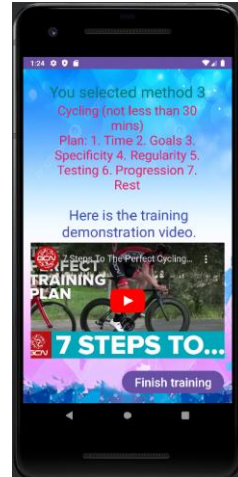
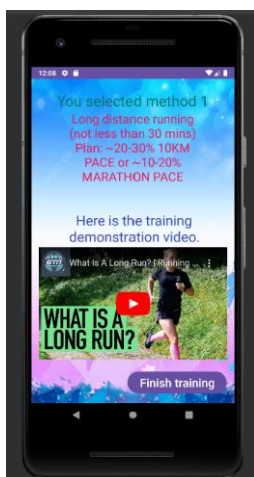
- The prototypes sample above draws 9 activities which are the application have
- If the application first launch, it would show the Home page and user can fill in his/her name
- Main page has 4 buttons: Start, Story, Setting and About Us, two cartoon characters can be touched and triggered the animation
- When press the “Start” button, it would turn to the Start page, user can select one training objective in five options. Then press “Next” button, it would turn to the Method page, user can select one training method in three options. Then press “Next” button, it would turn to the Training page, it would display the training plan and demonstration video. When finished training, press “Finish training” button, it would turn to the Finish page. It displays “Well done” for encouraging user. Then turn back to the main page
- In Story page, it displays the “Healthy Buddy Story” title and the content
- In Setting page, user can change the user’s name and turn on/off the background music
- In About Us page, it displays the content about myself
- Further detailed design focuses on relax interface and base on users’ training objective and interested method to decide the training method

**D) Implementation / Production / Testing** (Major production and testing issues only)

- During implementing and producing the mobile app (prototype), the major production is creating four buttons in the MainActivity and handling the function of StartActivity, MethodActivity, and TrainingActivity. I need to put the YouTube video in the TrainingActivity and link two selections to display the corresponding training page
- The coding if selected option 1 in StartActivity and the different result would display if selected different training method in MethodActivity:

```
if (startOption.equals("1") && methodOption.equals("1")) {
    method.setText("You selected method 1");
    name.setText("Long distance running (not less than 30 mins)\n"+
        "Plan: ~20-30% 10KM PACE or ~10-20% MARATHON PACE");
    videoUrl = "https://www.youtube.com/embed/Zo0nJU6k1CE";
}
else if (startOption.equals("1") && methodOption.equals("2")) {
    method.setText("You selected method 2");
    name.setText("Swimming (not less than 30 mins)\n"+
        "Plan: 1. Warm up 2. Pre-set 3. Main set 4. Post-set 5. Cool down");
    videoUrl = "https://www.youtube.com/embed/oM4sh1h1hTEE";
}
else if (startOption.equals("1") && methodOption.equals("3")) {
    method.setText("You selected method 3");
    name.setText("Cycling (not less than 30 mins)\n"+
        "Plan: 1. Time 2. Goals 3. Specificity 4. Regularity " +
        "5. Testing 6. Progression 7. Rest");
    videoUrl = "https://www.youtube.com/embed/il5vP83o1WM";
}
```

Testing result:



### 3) Development Environment, Evaluation and Testing

*Development Platforms and Tools:*

Android Studio (Windows Hedgehog | 2023.1.1)

Android Version: 6.0 API: 23 Code: Marshmallow

AVD version (and other devices) for testing: Pixel 2 API 28

*Errors and Warnings:*

**Major Known Errors** (not solved yet), with Descriptions and Remarks

Compile-time errors (and warnings), if any:

No Syntax Error found, but 11 Warning while compilation: 5 lines have "Not annotated parameter overrides @NonNull parameter", 1 line have "Method invocation 'getX' may produce 'NullPointerException'", 2 lines have "Anonymous new View.OnClickListener() can be replaced with lambda", 1 line have "Condition 'username != null' is always 'true'", 1 line have "Use 'SwitchCompat' from AppCompatActivity or 'SwitchMaterial' from Material library", 1 line have "Casting 'findViewById(...)' to 'Switch' is redundant", 1 line have "Anonymous new CompoundButton.OnCheckedChangeListener() can be replaced with lambda".

Run-time / Semantic errors during execution, if any:

No run-time error under normal operation.

#### 4) List of Additional Features or Techniques Applied if any

SharedPreferences – It is used in Activity (Home, Main, Setting) to share the user name

MediaPlayer - It is used in Activity (Main, Story, Setting, Us, Start, Method) to play the background music

GestureDetector - It is used in Activity (Main) to handle touch event

Animation - It is used in Activity (Main) to handle animation

#### 5) References

Background Music: <https://www.chosic.com/download-audio/29425/>

Radio Button: <https://www.geeksforgeeks.org/android-how-to-add-radio-buttons-in-an-android-application/>

Add video: <https://www.youtube.com/watch?v=V2KCAfHjySQ>

Touch and animation: <https://hiskio.com/courses/329/about?preview=15654>

~ END ~